Software Architectures for Big Data: Software Architectures from a Data Engineering Perspective

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Recall the general instructions shared in the previous exercise session

Exercise 2

- Link: https://github.com/ddibiasi/sbd-ais-ws26/tree/main/Exc_2
- Deadline: 21.10.2026
- Follow the modus operandi from the Github repository
- Add the link to the Github solution in the Moodle submission
- 2-3 people will be picked out every exercise block to present their work
- Be ready to shortly present (5–8 minutes) in the next exercise session (05.11.2026)

Resources

[1] THALPATI, Gaurav; **Practical Lakehouse Architecture: Designing and Implementing Modern Data Platforms at Scale**. O'Reilly Media, 2024.

[2] KLEPPMANN, Martin, and RICCOMINI, Chris; **Designing data-intensive applications: The big ideas behind reliable, scalable, and maintainable systems**. 2nd ed. O'Reilly Media, 2026.

[3] DULAY, Hubert, and DEBUSMANN, Ralph. **Streaming Databases: Unifying Batch and Stream Processing**. O'Reilly Media, 2024.

[4] PETROV, Alex. **Database Internals: A deep dive into how distributed data systems work**. O'Reilly Media, 2019.

[5] VACCA, John; **Cloud computing security: foundations and challenges**. CRC Press, 2016.

[6] REIS, Joe, and HOUSLEY, Matt. **Fundamentals of Data Engineering: Plan and Build Robust Data Systems**. O'Reilly Media, 2022.

[7] HUYEN, Chip. **Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications**. O'Reilly Media, 2022.

[8] RICHARDS & FORD, **Fundamentals of Software Architecture**, O'Reilly Media, 2020.